

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical
Practical – X

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a. Create a PWA With a Web App manifest file Including.

Code(manifest.json):-

```
{
  "name": "My PWA Application",
  "short_name": "MyPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "447x447",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "447x447",
      "type": "image/png"
    }
  ]
}
```

Code(index.html):-

```
<!DOCTYPE html>
<html>
<head>
  <title>My PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#0d6efd">
</head>
<body>
  <h1>My Progressive Web App</h1>
  <button onclick="showNotification()">
    Show Notification
  </button>
  <script src="app.js"></script>
</body>
</html>
```

Code(app.js):-

```
if ('serviceWorker' in navigator)
{
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker Registered'));
}
function showNotification() {
}
```

```
Notification.requestPermission().then(permission =>
{
  if (permission === 'granted')
  {
    navigator.serviceWorker.ready.then(registration =>
    {
      registration.showNotification('PWA Notification', {
        body: 'Push Notification Working Successfully!',
        icon: 'icons/icon1.png'
      });
    });
  }
});
```

Code(service-worker.js):-

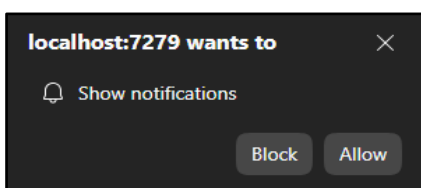
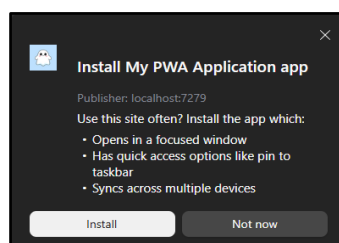
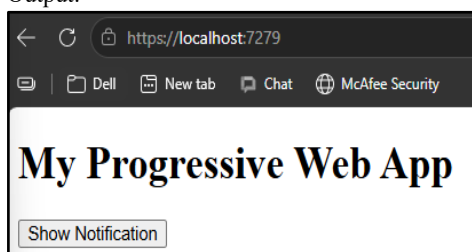
```
self.addEventListener('install', event =>
{
  console.log('Service Worker Installed');
});

self.addEventListener('activate', event =>
{
  console.log('Service Worker Activated');
});
```

Code(program.cs):-

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

Output:-



b.Develop an E-Commerce PWA with product listing and offline cart functionality.

Code(program.cs):-

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

Code(index.html):-

```
<!DOCTYPE html>
<html>
<head>
  <title>E-Commerce PWA</title>
```

```
<link rel="manifest" href="manifest.json">
<meta name="theme-color" content="#0466d6">
<style>
  body {
    font-family: Arial;
    margin: 20px;
  }
  .container {
    display: flex;
    gap: 30px;
  }
  .cart-section {
    width: 30%;
    border: 1px solid gray;
    padding: 15px;
    min-height: 300px;
  }
  .product-section {
    width: 70%;
  }
  .products {
    display: flex;
    flex-wrap: wrap;
    gap: 15px;
  }
  .product {
    border: 1px solid gray;
    padding: 10px;
    width: 200px;
  }
  button {
    padding: 5px 10px;
    cursor: pointer;
  }
</style>
</head>
<body>
  <h1>E-Commerce PWA</h1>
  <div class="container">
    <!-- Cart Section -->
    <div class="cart-section">
      <h2>Cart Items</h2>
      <ul id="cartList"></ul>
    </div>
    <!-- Product Section -->
    <div class="product-section">
      <h2>Products</h2>
      <div class="product">
        <h3>Laptop</h3>
        <p>Price: $ 5000</p>
        <button onclick="addToCart('Laptop')">
          Add to Cart
        </button>
      </div>
      <div class="product">
```

```

        <h3>Mobile</h3>
        <p>Price: $ 2000</p>
        <button onclick="addToCart('Mobile')">
            Add to Cart
        </button>
    </div>
    <div class="product">
        <h3>Headphones</h3>
        <p>Price: $ 3000</p>
        <button onclick="addToCart('Headphones')">
            Add to Cart
        </button>
    </div>
</div>
<script src="app.js"></script>
</body>
</html>

```

Code(app.js):-

```

if ('serviceWorker' in navigator)
{
    navigator.serviceWorker.register('service-worker.js')
        .then(() => console.log('Service Worker Registered'));
}
let cart = JSON.parse(localStorage.getItem('cart')) || [];
displayCart();
function addToCart(product)
{
    cart.push(product);
    localStorage.setItem('cart', JSON.stringify(cart));
    displayCart();
}
function displayCart()
{
    let cartList = document.getElementById('cartList');
    cartList.innerHTML = "";
    cart.forEach(item => {
        let li = document.createElement('li');
        li.textContent = item;
        cartList.appendChild(li);
    });
}

```

Code(manifest.json):-

```

{
    "name": "E-Commerce PWA",
    "short_name": "ShopPWA",
    "start_url": "/index.html",
    "display": "standalone",
    "background_color": "#ffffff",
    "theme_color": "#0d6efd",
    "icons": [
        {
            "src": "icons/icon1.png",
            "sizes": "447x447",
            "type": "image/png"
        }
    ],
}

```

```

{
  "src": "icons/icon2.png",
  "sizes": "447x447",
  "type": "image/png"
}
]
}
}

```

Code(service-worker.js):-

```

const CACHE_NAME = 'ecommerce-cache-v1';
const urlsToCache = [
  '/',
  '/index.html',
  '/app.js',
  '/manifest.json'
];

self.addEventListener('install', event => {
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(cache => {
        return cache.addAll(urlsToCache);
      })
  );
});

self.addEventListener('fetch', event => {
  event.respondWith(
    caches.match(event.request)
      .then(response => {
        return response || fetch(event.request);
      })
  );
});

```

Output:-

The top screenshot shows the E-Commerce PWA interface. It has a header 'E-Commerce PWA' and two main sections: 'Cart Items' and 'Products'. The 'Cart Items' section lists 'Laptop', 'Mobile', and 'Headphones'. The 'Products' section lists 'Laptop', 'Mobile', and 'Headphones', each with a price and an 'Add to Cart' button.

The bottom screenshot shows the same interface with the 'Network' tab open in the developer tools. The 'Network' tab displays a list of network requests, including 'localhost' and 'browserLink'. The 'Elements' panel shows the 'Laptop' product in the 'Products' section.